

Team name: Pipe Dream

Team members:

- Damien Akers
- Joshua Pepple
- Marley Williams
- Saef Al Barazi

Represented institution: University of the West of England

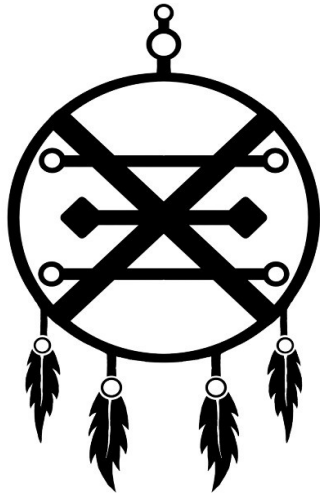
Main design parameters/description:

- using weight offset to cling onto the pipe
- 3D printed PLA for chassis and housing
- driven by a single 12v DC motor with an rpm of 200
- Arduino nano for logic control

Competition parameters/goal:

- Climb a 22mm diameter copper pipe until reaching the top datum surface
- Do so whilst carrying a chain, dropping it at the midway point of the pipe
- Shine a Green LED when the robot is active
- Activate a red LED and a buzzer when top datum surface is reached

Team Logo



Design process:
Initial goal of 10s travelling time. Rough bottom to top time needs to be 3.33s
 $F = ma$, assuming $m = 1.2\text{kg}$ and $a = 9.81$.
 $F = 11.8$
 $W = F \cdot d$, $d = 1.8$
 $W = 21\text{j}$
 $P = W/t$
Req power = 6.5W, with 1.5FOS, roughly 10W.

Information:

- CAD mass: 787 grams not counting Arduino and h-bridge
- 12v DC motor with worm gears for self-locking
- Arduino Nano
- Dual H bridge motor XY-160D
- Total cost, including consumables: £78.91

Noteworthy Electronics Components

Component	Role	Selection Justification
Arduino Nano R4	Central Logic	Lightweight, High-speed
XY-160D H-Bridge	Motor Driver	High current capacity
Worm Gear Motor	Primary Actuator	Self-locking to prevent sliding
Optical Encoder	Distance Feedback	Provides precision tick for finding distance
7809CT Regulator	Power Management	Stabilises the 18v battery input into 9V for arduino

